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Unidad 1

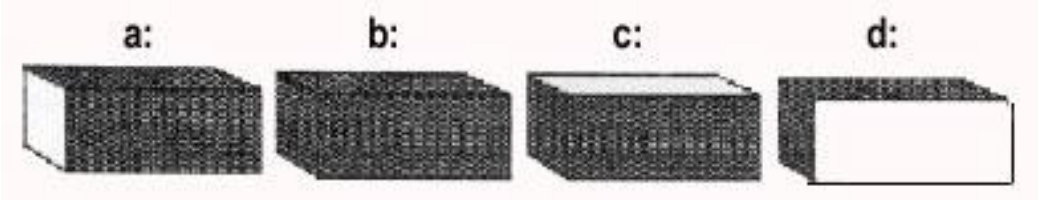
Razonamiento Espacial

El razonamiento espacial evalúa la capacidad del individuo para visualizar objetos en su mente, así como la habilidad de imaginar un objeto en diferentes posiciones, sin perder de él sus características, como por ejemplo, la rotación de imágenes o la construcción de figuras; también se incluyen las habilidades para descubrir similitudes (semejanzas) entre objetos que parecen diferentes.
Esta capacidad de percibir correctamente el espacio, sirve para orientarse mediante planos y mapas y le permite al ser humano crear dibujos, construir estructuras en tres dimensiones (3D), tales como esculturas, edificios, etc.
La noción de "imagen" juega un papel importante en el estudio de la habilidad espacial. El razonamiento espacial muestra la habilidad de una persona para visualizar la forma y las superficies de un objeto terminado, antes de ser construido.

Ejercicio 1

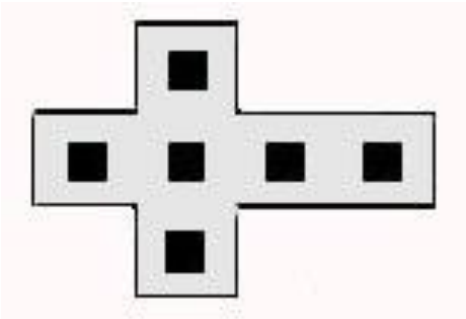
Cuál de las 4 figuras (a, b, c, d) se puede armar al doblar el modelo siguiente:

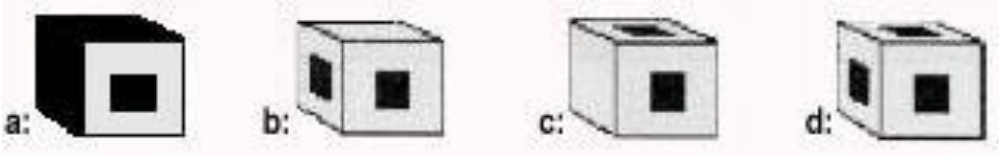




Como el modelo del ejemplo es totalmente oscuro, solamente se podrá armar una "figura completamente oscura" al doblar dicho modelo.
Por lo tanto, la respuesta será la indicada con la letra "b", ya que las otras figuras tienen sectores blancos.

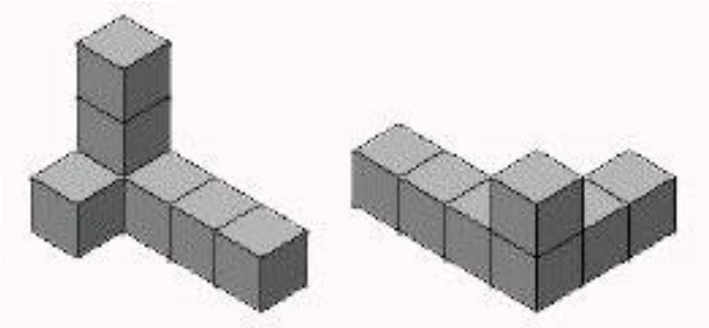
Ejercicio 2
Cuál de las 4 figuras (a, b, c, d) se puede armar al doblar el modelo:





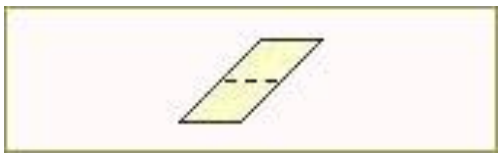
Como el modelo tiene un cuadro negro en cada uno de sus lados, sólo se podrá armar una figura que tenga "cuadros negros en cada uno de sus lados".
Solamente la respuesta "d" tiene una figura con esas características.

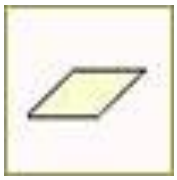
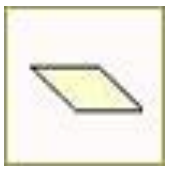
Ejercicio 3
Al sobreponer (superponer) las dos figuras, ¿quedan exactamente iguales?



Sí – No
Aunque ambas figuras tienen el mismo número de cubos, NO encajan exactamente, porque dichos cubos están ubicados en diferente posición.

Ejercicio 4
Si doblamos la figura por la línea punteada, ¿qué forma (a, b, c) resultará?



a:  b:  c: 

La figura "a" tiene la misma forma que cualquiera de las mitades de la figura modelo.

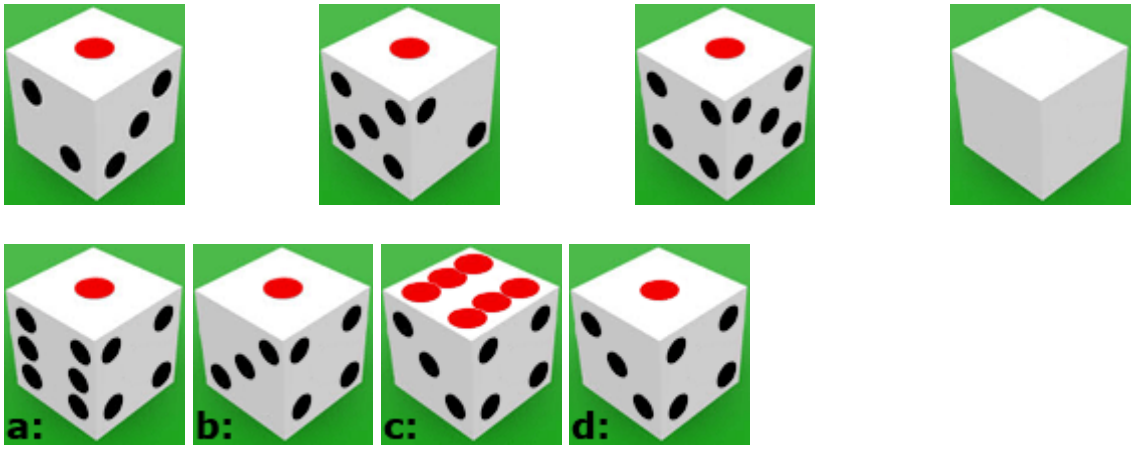
Ejercicio 5
¿Cuál de las figuras NO pertenece al grupo?

a:  b:  c:  d:  e: 

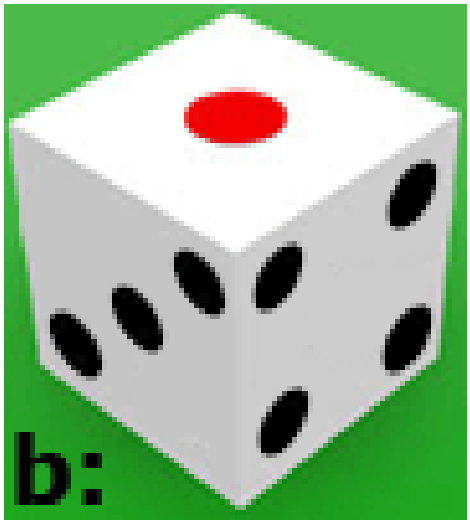
Si sobreponemos (superponemos) las figuras, la marcada con la "d" NO encaja con las demás.

Ejercicio 6

¿Cuál de las figuras (a, b, c, d), continúa la secuencia?



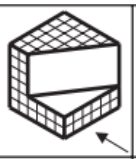
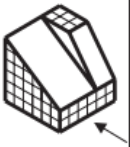
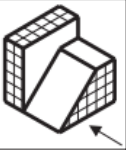
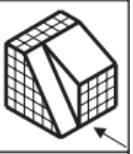
El dado gira horizontalmente, por sus 4 caras posibles.
La respuesta es **"b"**.



Perspectivas de objetos

Esta perspectiva representará al objeto en su posición de trabajo, y en caso de que pueda ser utilizable en cualquier posición, se representará en la posición de mecanizado o montaje.
En ocasiones, el concepto anterior puede no ser suficiente para elegir el alzado de una pieza, en estos casos se tendrá en cuenta los principios siguientes:

1. Conseguir el mejor aprovechamiento de la superficie del dibujo.
2. Que el alzado elegido, presente el menor número posible de aristas ocultas.
3. Y que nos permita la obtención del resto de vistas, planta y perfiles, lo más simplificadas posibles.

						
ALZADO						
PLANTA						
PERFIL						


1


2


3


4


5


6


7


8


9


10


11


12


13

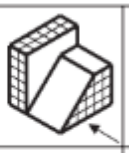
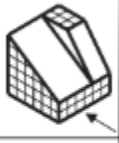
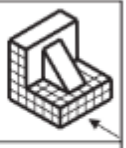

14

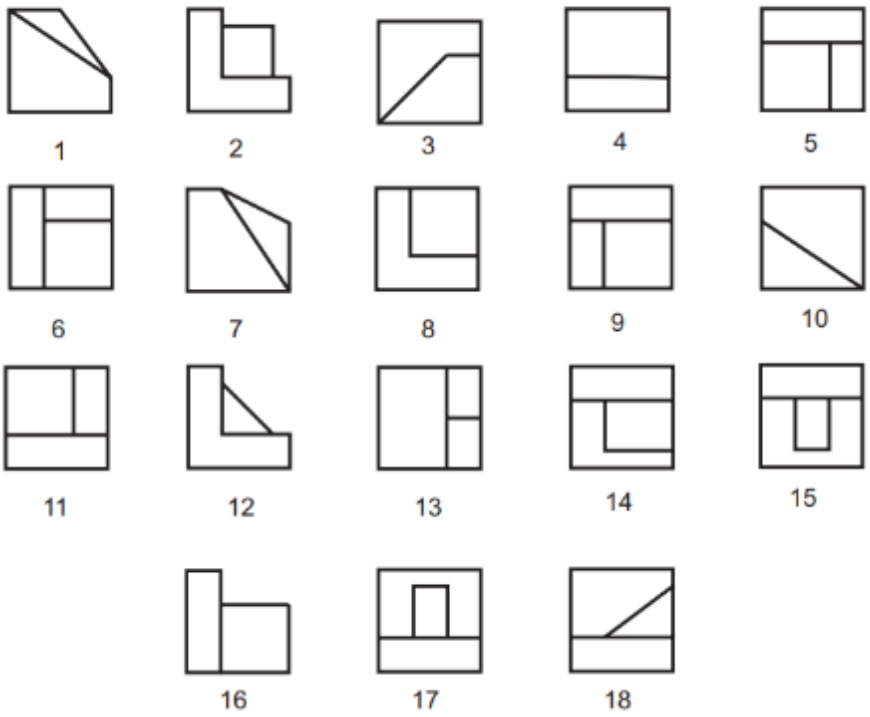

15


16


17


18

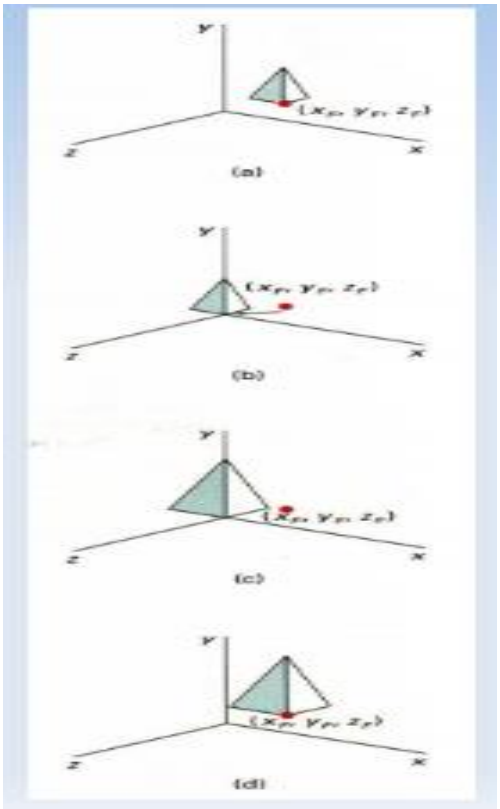
						
ALZADO						
PLANTA						
PERFIL						

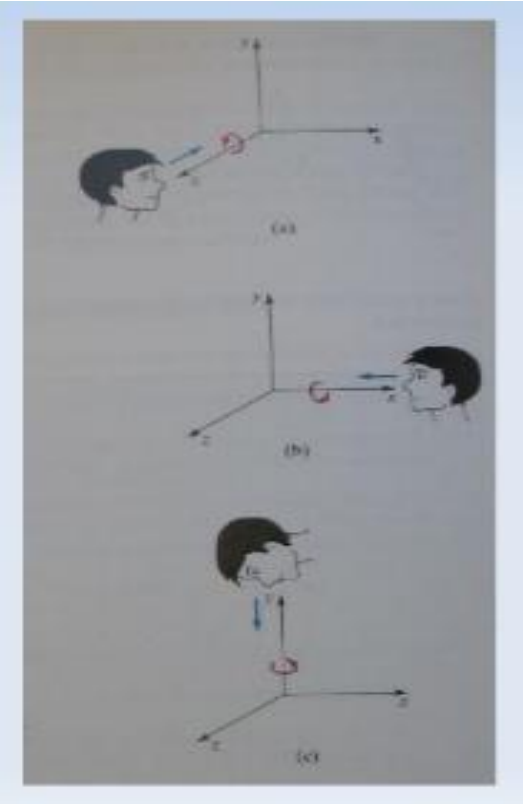


Transformación grafica de 3D y 2D

Representación Gráfica 3D y 2D de objetos reales o imaginarios a partir de modelos matemático. Se usan algoritmos, procedimientos que permiten obtener la imagen master final en 2D Procesando imágenes a un proceso inverso a 3D, estas son Analizadas con imágenes **2D, o reconstrucción de modelos 2D y 3D a partir de sus imágenes.**

Ejemplo:





2D **3D**

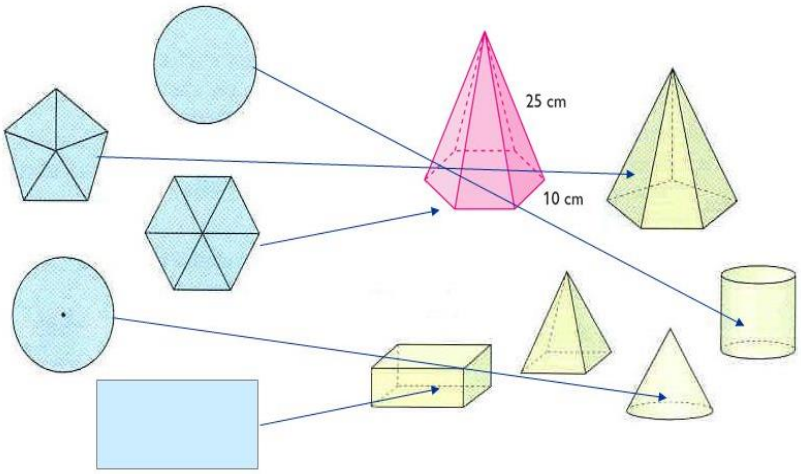


Figura rotativa

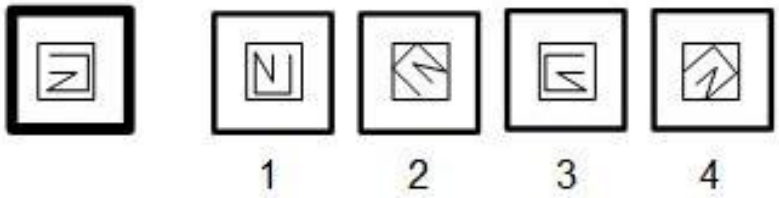
Figura rotativa Es un ejercicio que permite determinar la capacidad de estructurar y comprender las relaciones espaciales, que resultan básicas en importantes áreas de conocimiento como son la Geometría, la Arquitectura y la Ingeniería.

Ejemplo:

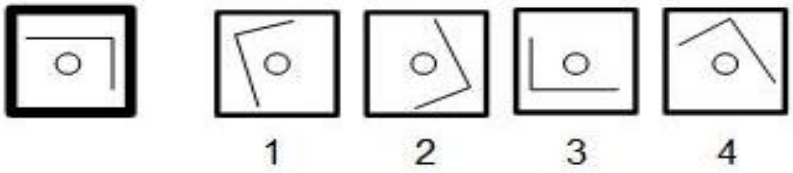
Ejercicios de Rotación de Figuras:

-¿Cuál es la figura **distinta** a la del modelo?:

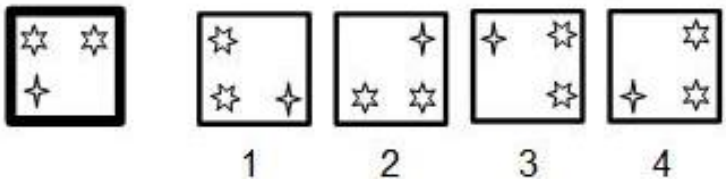
1-



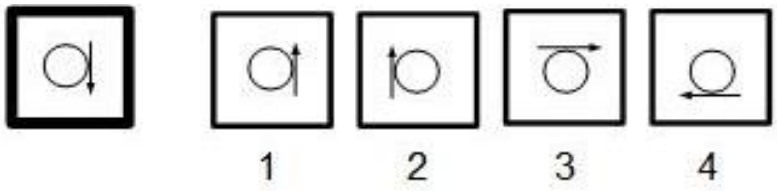
2-



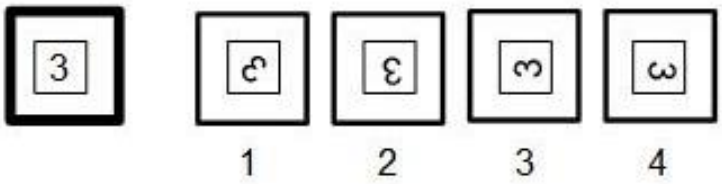
3-



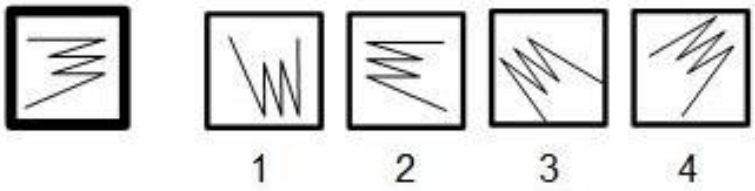
4-



5-

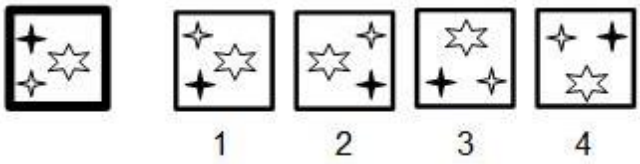


6-

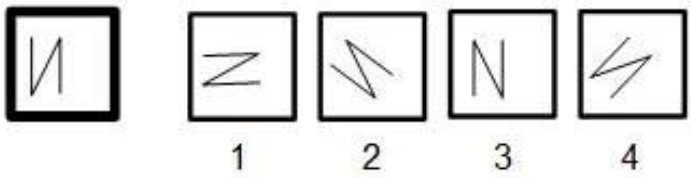


- ¿Cuáles son las figuras **iguales** al modelo?

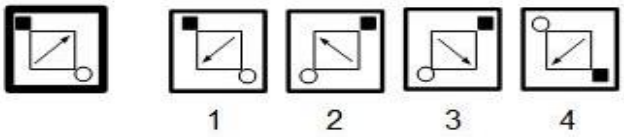
7-



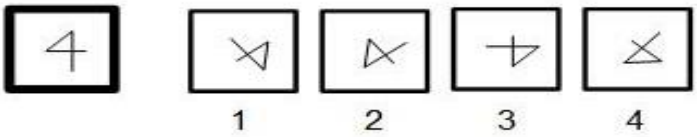
8-



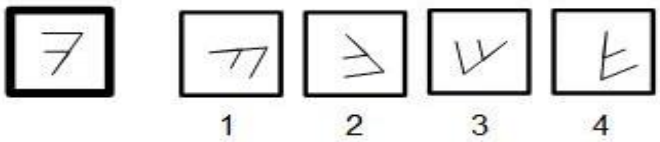
9-



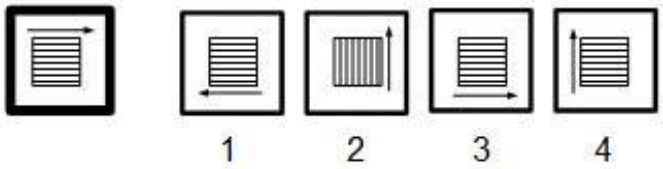
10-



11-



12-



INTERPRETACIÓN DEL PATRÓN PARA COMPLETAR LA SERIE GRÁFICA

Un patrón es una sucesión de signos orales, de fenómenos naturales, gráficos, numéricos, entre otros, que se construyen siguiendo una regla, ya sea de repetición o de recurrencia.9 Los patrones se observan en la vida real y pueden responder a un modelo matemático.

Patrones numéricos Están formados por sucesión de números y operaciones escritos en un orden definido. Por ejemplo: La secuencia de este patrón está formada por números enteros y decimales. También son patrones numéricos las coordenadas de un plano cartesiano. Por ejemplo:

1, 1.5, 2, 2.5, 3, 3.5 ...

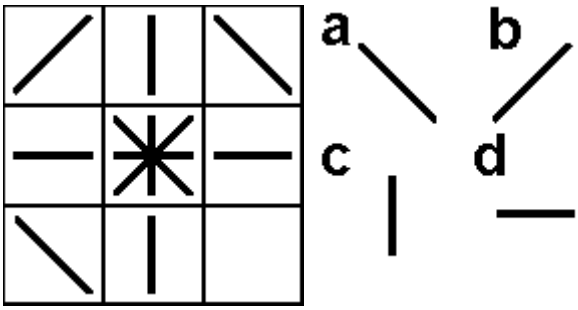
También son patrones numéricos las coordenadas de un plano cartesiano. Por ejemplo:



EJEMPLOS:

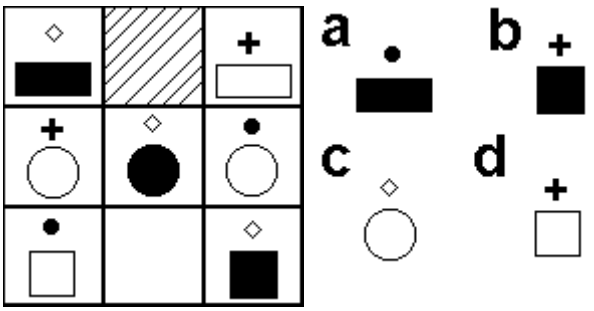
Rellenar la casilla vacía a la izquierda con la figura más lógica (a b c d) de la derecha.

1.



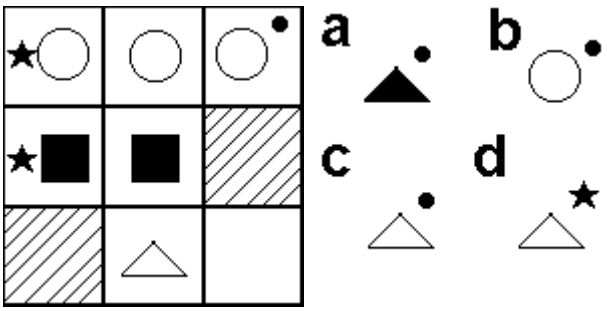
○ a ○ b ○ c ○ d

2.



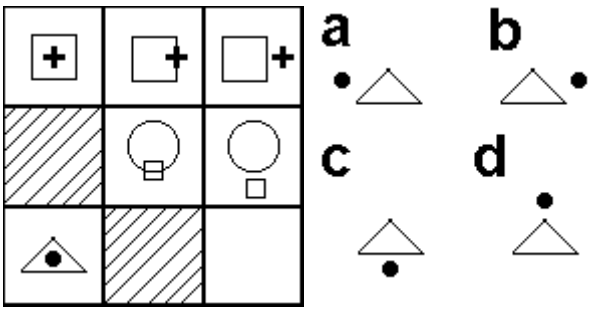
a b c d

3.



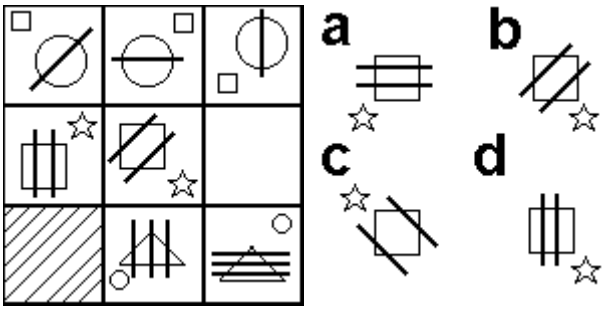
a b c d

4.

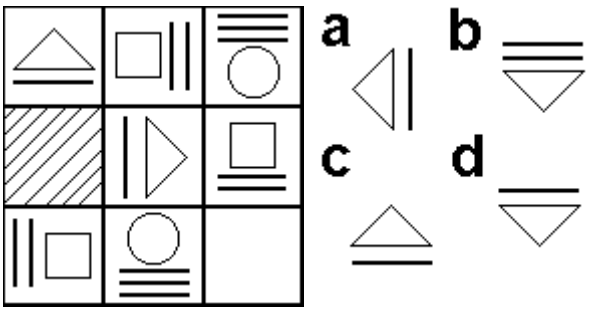


a b c d

5.

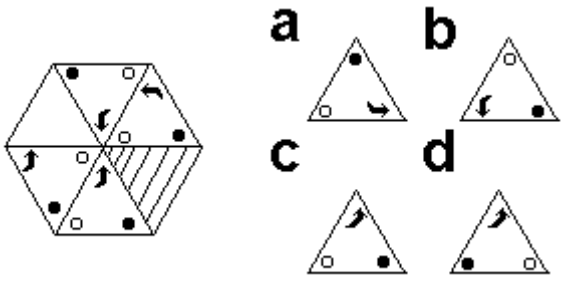


6.



a b c d

7.



a b c d

De las cuatro figuras de la derecha (a b c d) elegir la figura que continúa con más lógica tras las tres figuras de la izquierda

A | B | C | D

8.



a b c d

9.



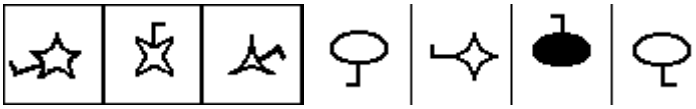
a b c d

10.



a b c d

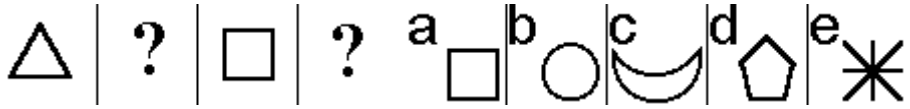
11.



a b c d

En cada línea faltan dos de las cuatro figuras (?). De las figuras de la derecha (a b c d e), elegir las dos que corresponden con más lógica.

12.



a y d b y d a y b c y a d y a

13.



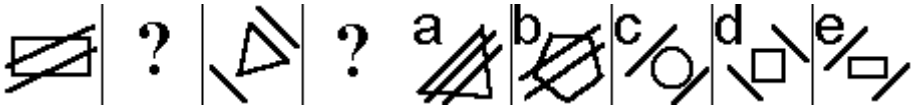
b y c a y d a y e b y a a y e

14.



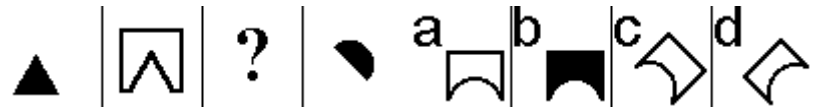
c y b a y b b y c a y d c y e

15.



a y b b y e b y c e y b e y c

16.



a b c d

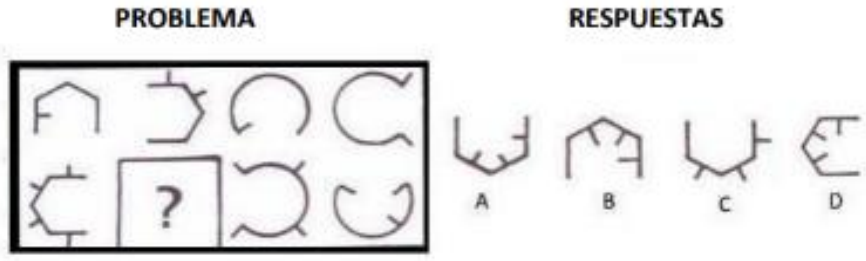
Unidad 2

SERIES GRAFICAS

COMPILACION:

Son continuaciones de series de figuras

De acuerdo con las figuras, ¿qué serie continua?

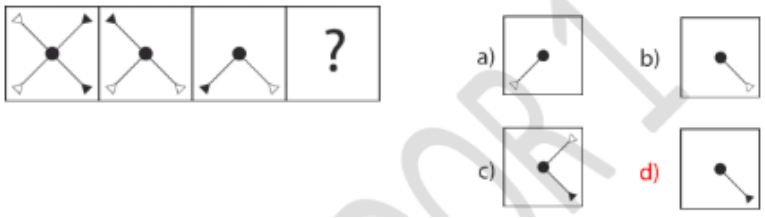


La primera fila tiene 1 y 2 rayas en la casita parada y acostada, luego 1 y 2 rayas en el arco parado y acostado. En la segunda fila tenemos 4 y 3 rayas en el arco acostado y de cabeza, y por lógica tendríamos 4 y 3 rayas en la casita acostada y de cabeza que sería la respuesta “a”.

SIMPLIFICACIÓN:

Son continuaciones de secuencias de figuras

De acuerdo con las figuras, ¿qué secuencia continua?



1. La primera figura tiene 4 segmentos de línea recta que se unen en el punto central, dos de ellos con conos negros y dos con conos blancos, formando una equis.

2. En la segunda la equis gira en contra de las manecillas del reloj en 90° y desaparece el segmento que va desde el centro hacia el vértice superior derecho del cuadro

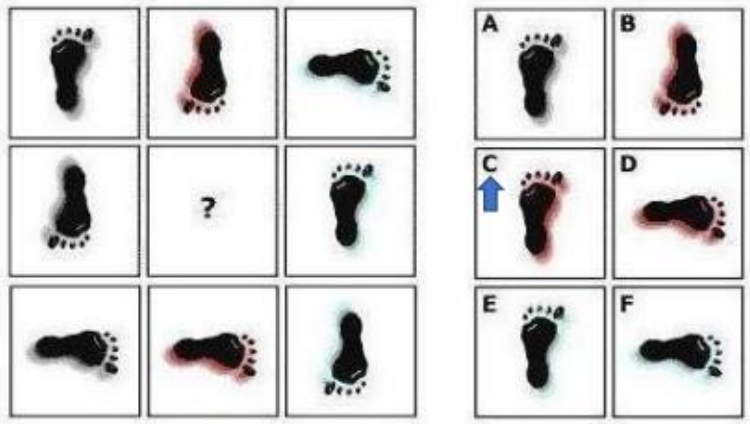
3. En la tercera, vuelve a girar en 90° en contra de las manecillas del reloj y desaparece el segmento que va desde el centro hacia el vértice superior derecho del cuadro

4. En la cuarta, vuelve a girar en 90° en contra de las manecillas del reloj y desaparece el segmento que va desde el centro hacia el vértice superior derecho del cuadro; por tanto, desaparece el segmento de cono blanco y nos queda el segmento de cono negro y la respuesta es el literal d).

Cambio posicional de figuras

Ejemplo:

Señala la posición correcta del píe en el recuadro que contiene la incógnita.



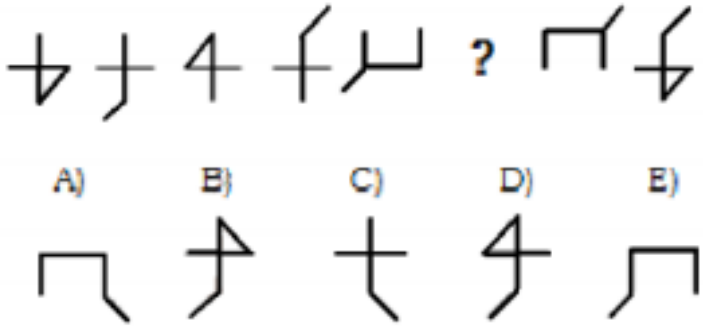
Arriba -abajo -derecha

Abajo – arriba – derecha

Derecha – arriba – abajo

Deformación de objetos

La deformación de objetos Infiere cuál es la figura que completa la serie.



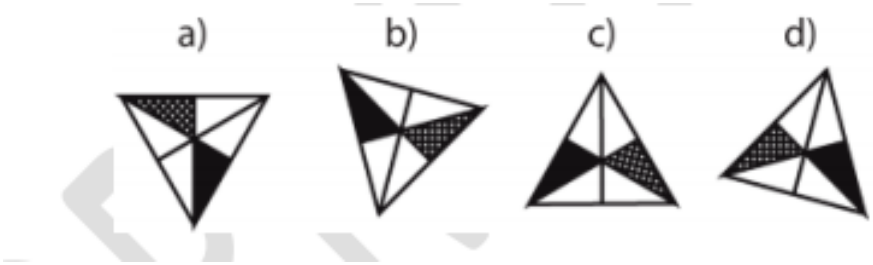
Unidad 3

Conjuntos gráficos

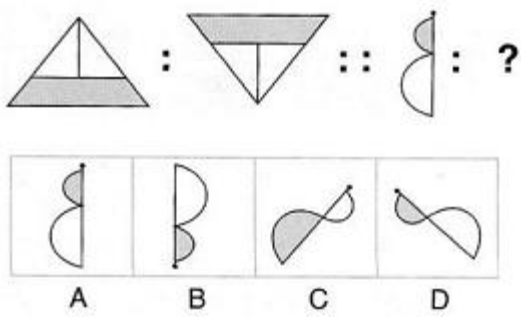
Semejanzas y diferencias graficas

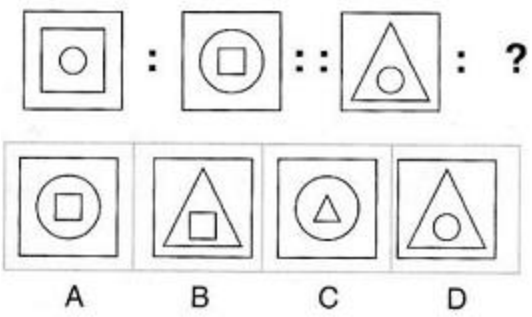
Ejemplo:

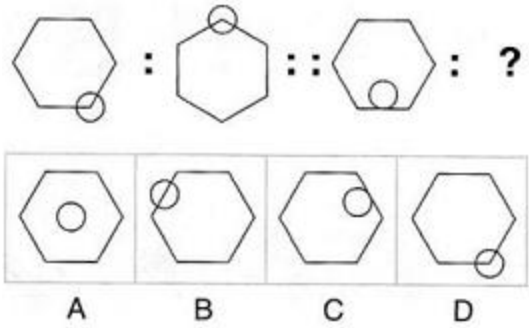
Las figuras perteneces al mismo conjunto, excepto

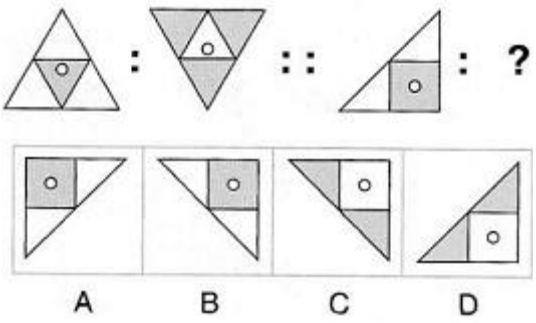


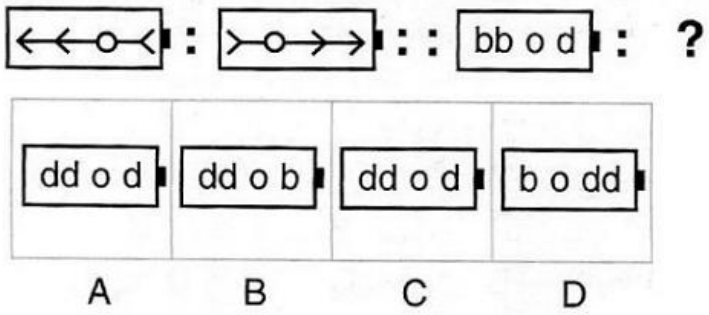
Analogías Gráficas



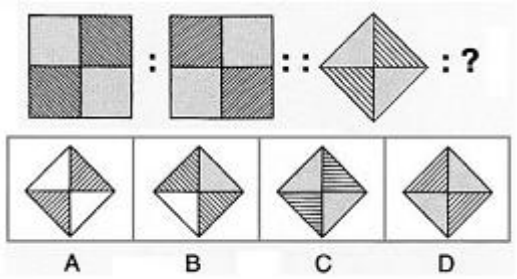


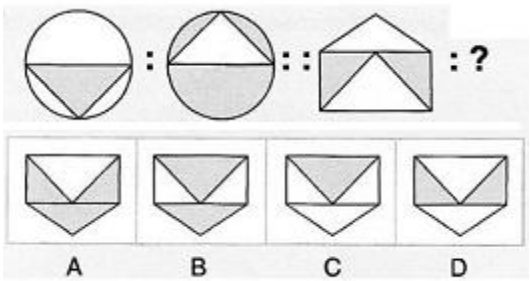


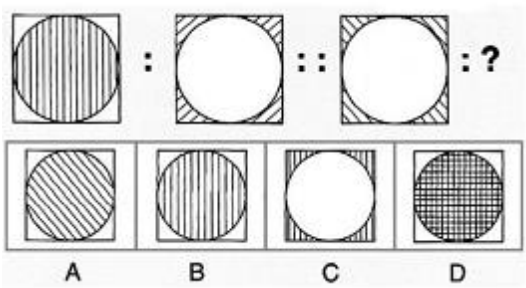


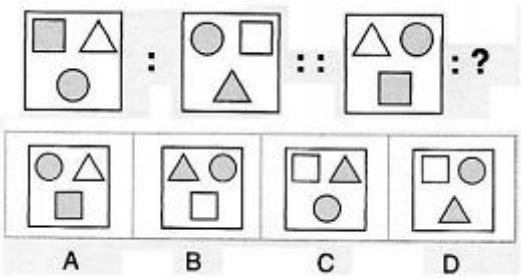


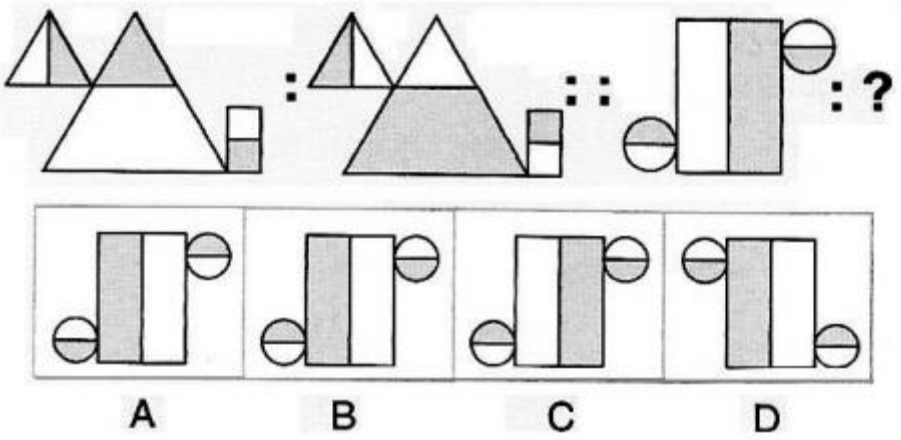
Ejercicios prácticos

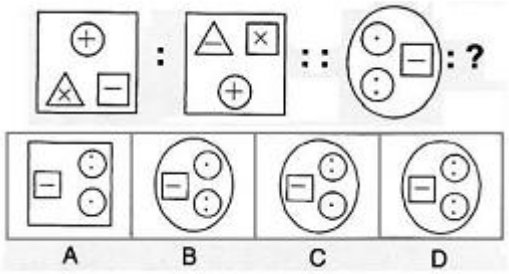


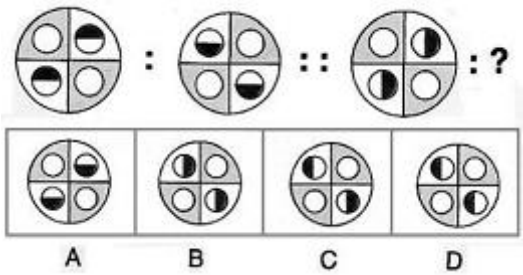


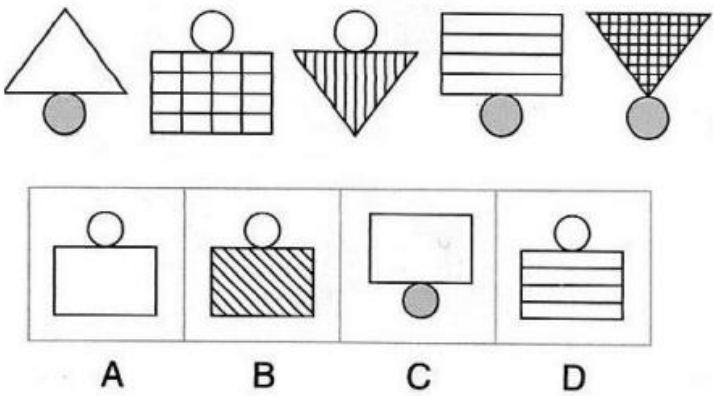


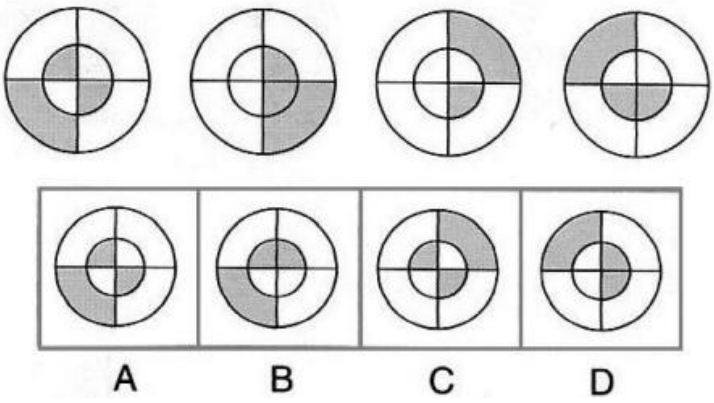


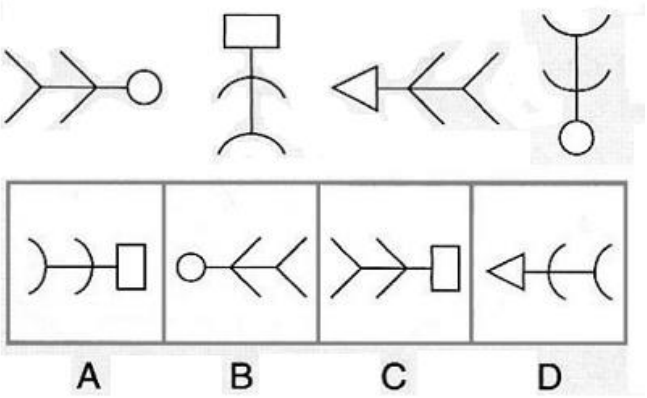

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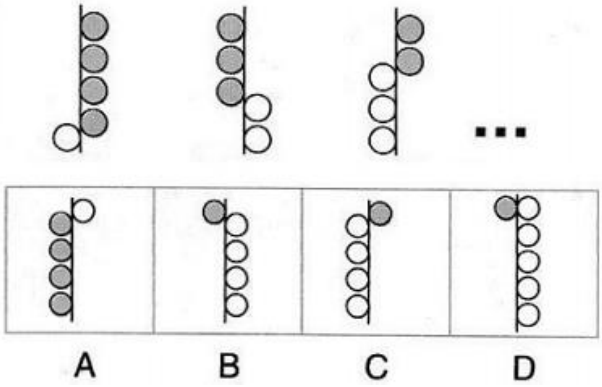

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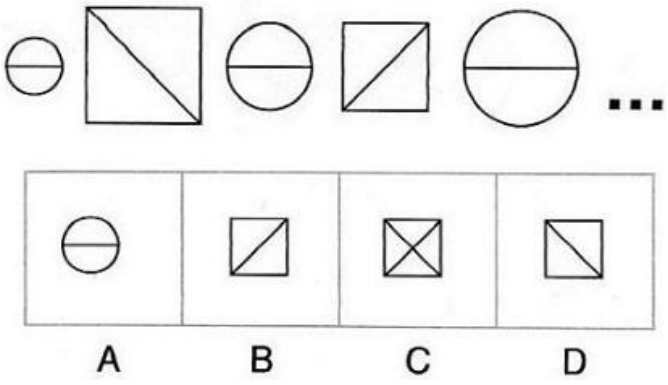

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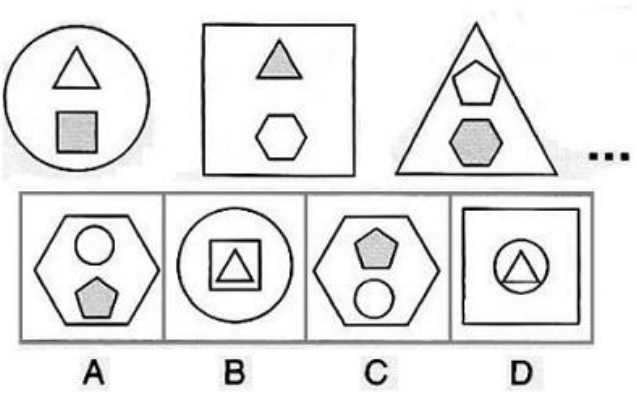

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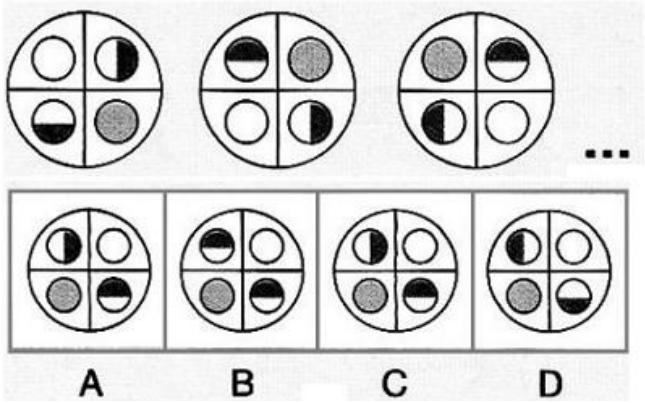

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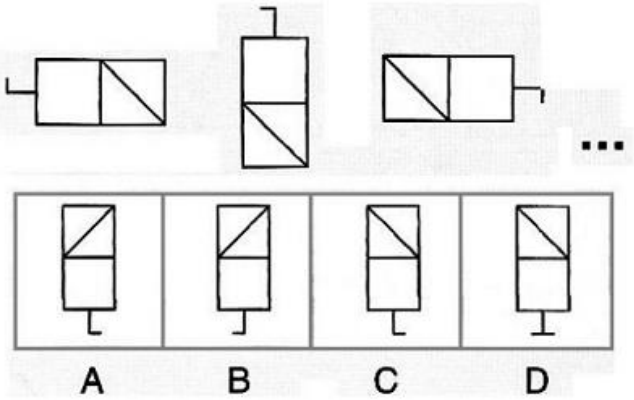

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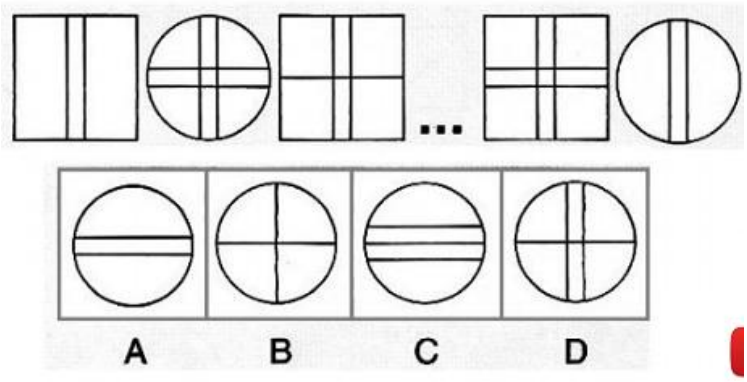

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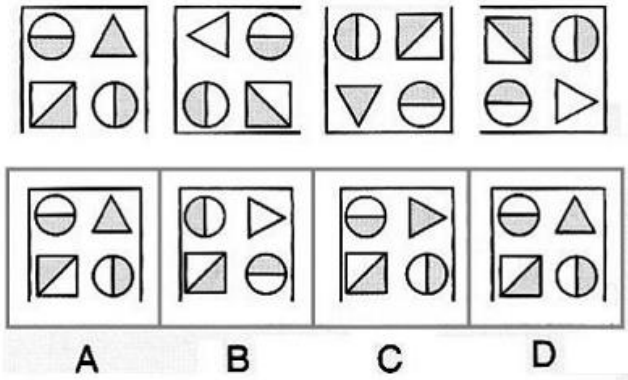

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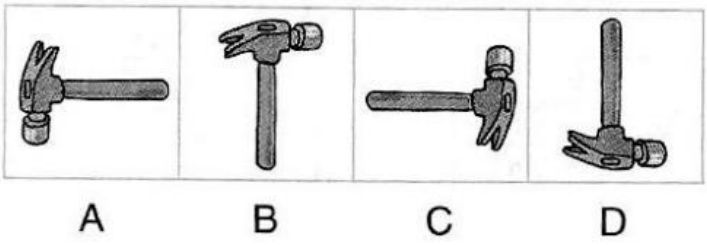

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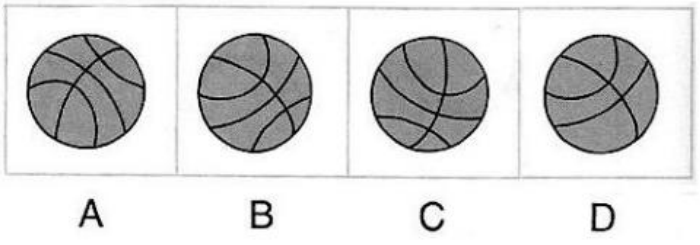

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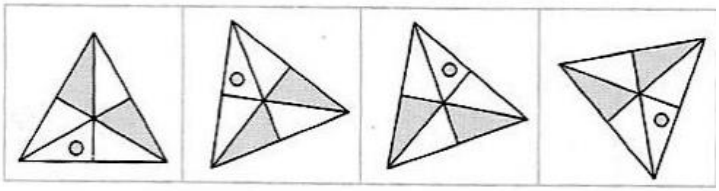

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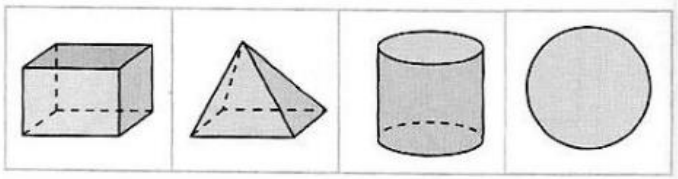
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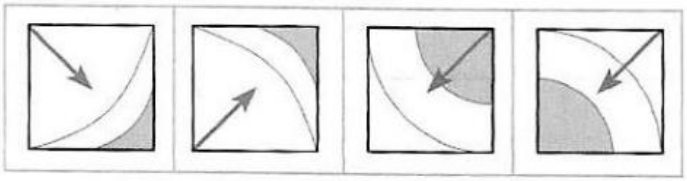

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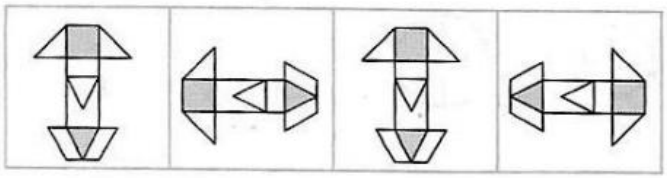
A B C D



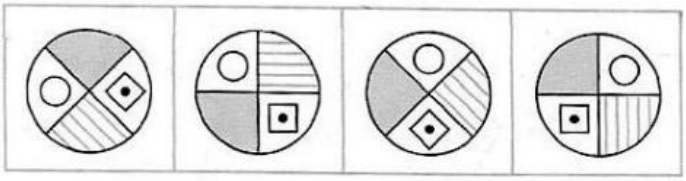
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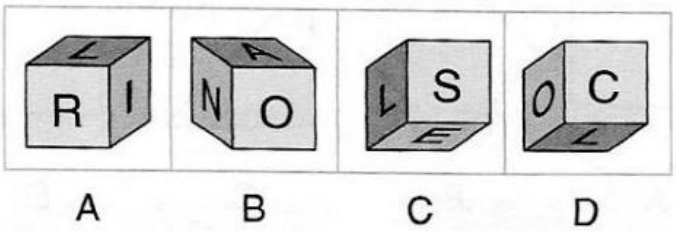
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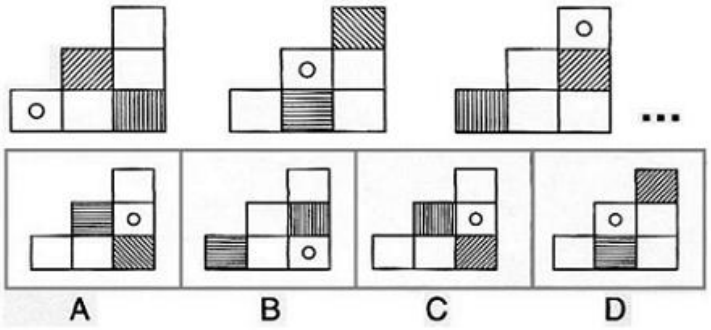


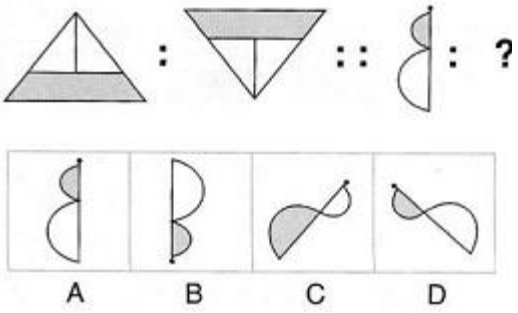
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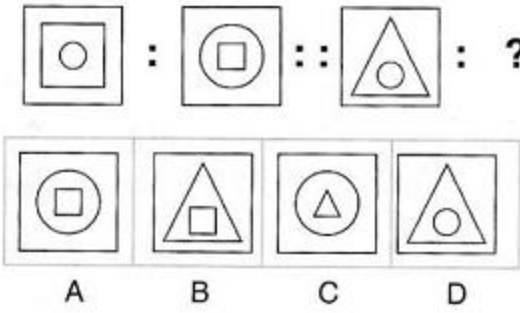


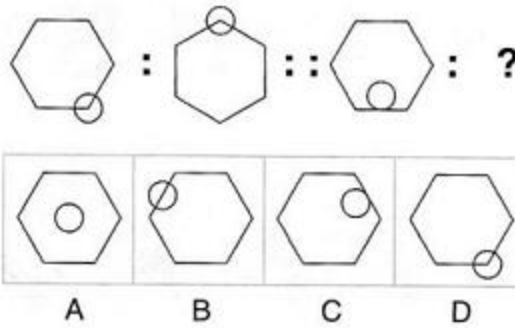
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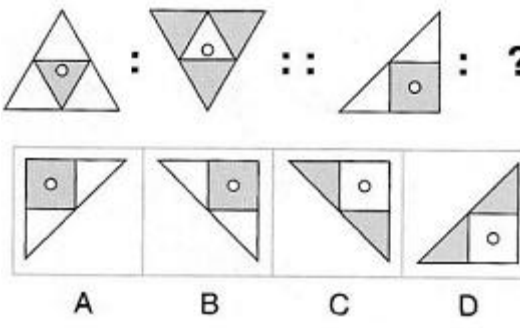

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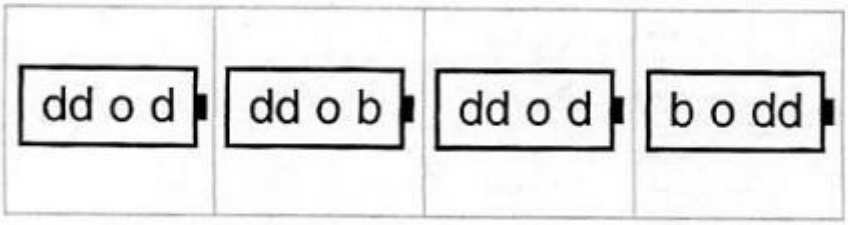

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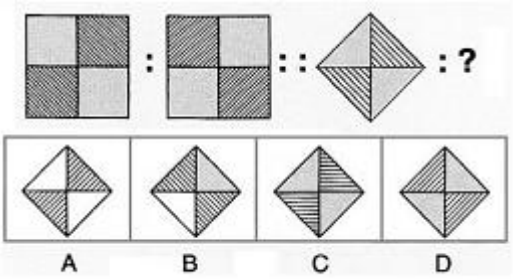

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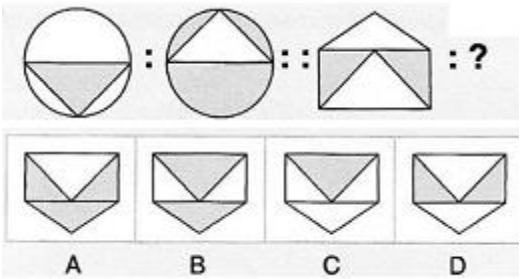

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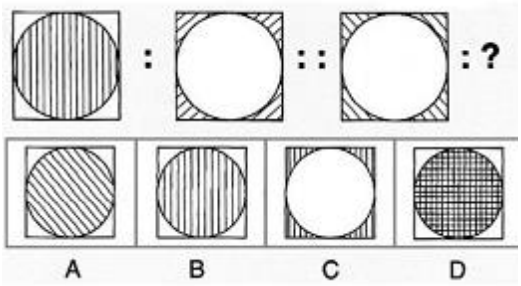

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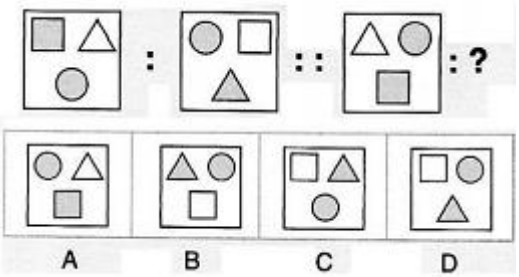

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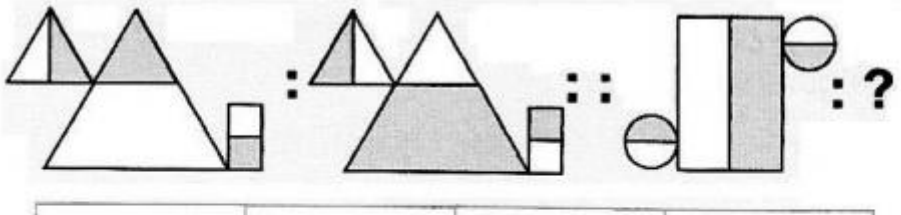

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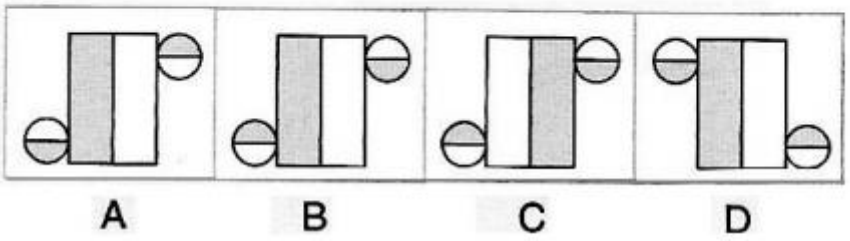

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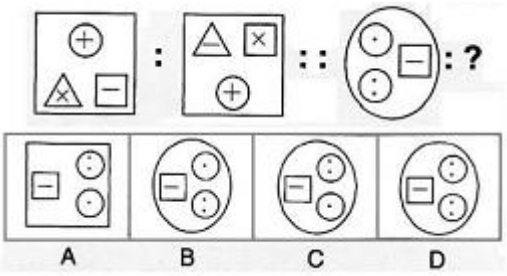

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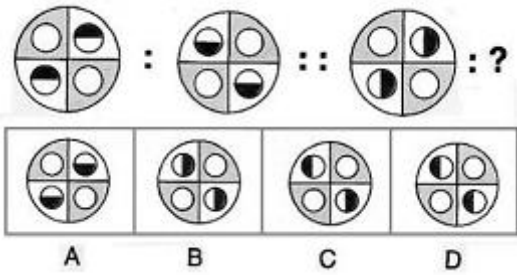

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D




A
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C
D





1. 1, 4, 10, 19, ?
   A 35 B 31 C 29

2. 6, 3, 12, 6, ?
   A 1 B 22 C 18

3. 1, 1, 4, 4, ?
   A 5 B 7 C 9

4. 2, 8, 3, 9, ?
   A 5 B 4 C 10

5. 
   A  B  C 

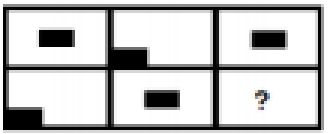
6. 
   A  B  C 

7. 
   A  B  C 

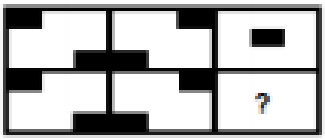
8. 8, 64, 512, 4096, ?
   A 32765 B 32768 C 32767

9. 3, 2, 9, 4, ?
   A 6 B 28 C 81

10. 1, 2, 4, 8, ?
    A 12 B 16 C 10

11. 
    A  B  C 

12.



A 

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C 

13.

A	D	G
C	F	?

A E

B I

C J

14.

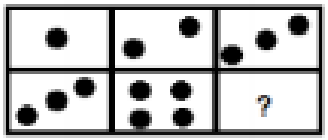
A1	D4	G7
C3	F6	?

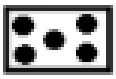
A I9

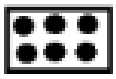
B E7

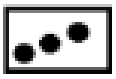
C J9

15.



A 

B 

C 

16.

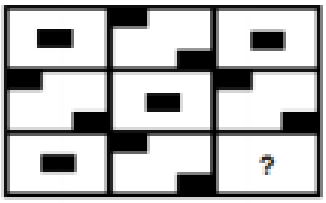
A	F	K
C	H	M
E	J	?

A O

B P

C M

17.

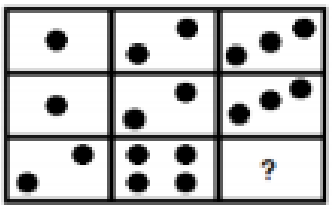


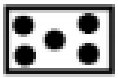
A 

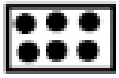
B 

C 

18.

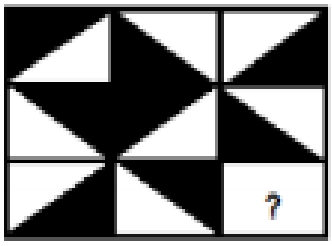


A 

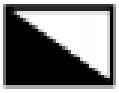
B 

C 

19.

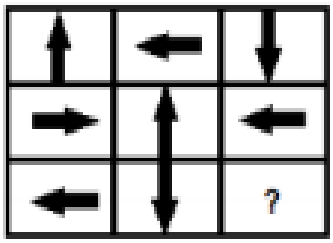


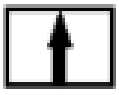
A 

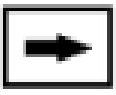
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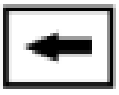
C 

20.

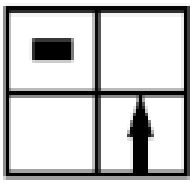


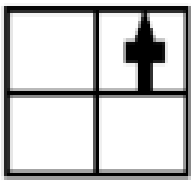
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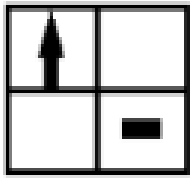
B 

C 

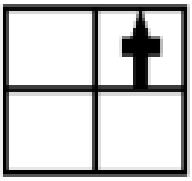
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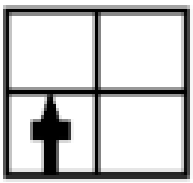


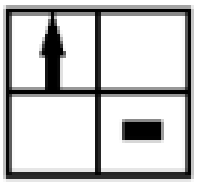




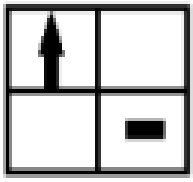


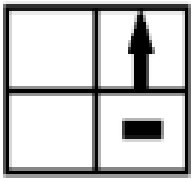
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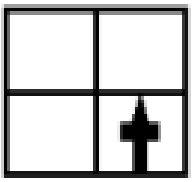
B 

C 

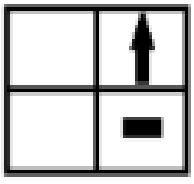
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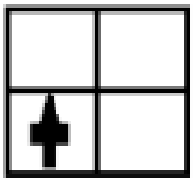


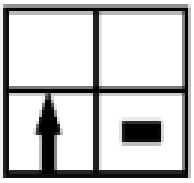


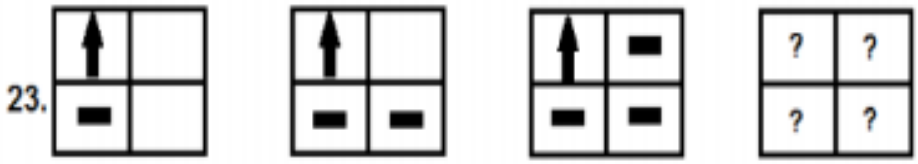


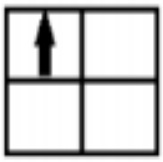
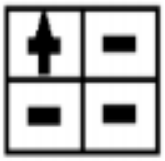
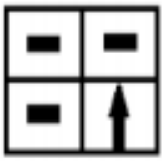


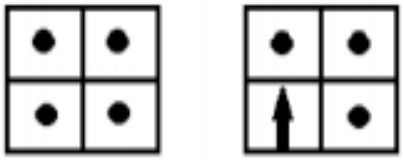
A 

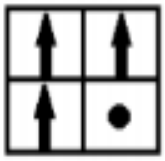
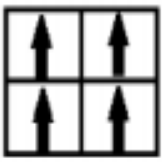
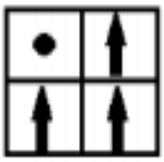
B 

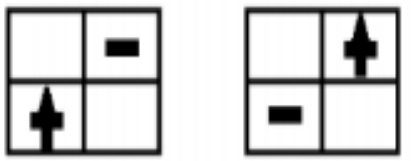
C 

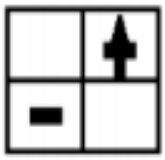
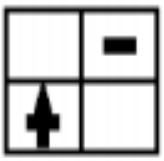
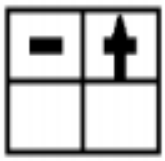
23. 

A  B  C 

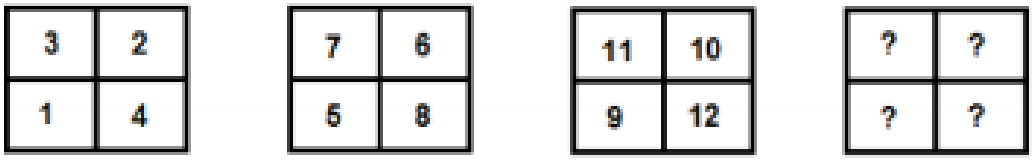
24. 

A  B  C 

25. 

A  B  C 

26.



A

14	15
13	16

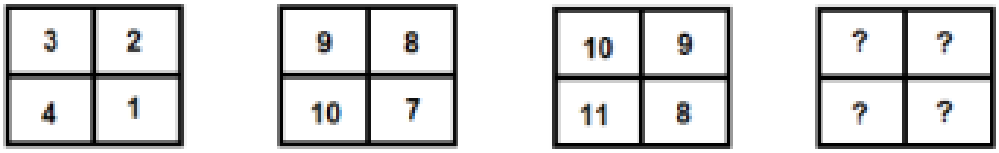
B

15	14
13	16

C

15	16
13	14

27.



A

6	7
5	8

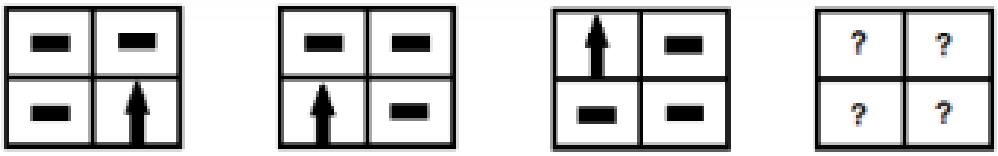
B

4	3
5	2

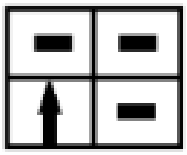
C

9	10
8	11

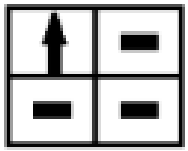
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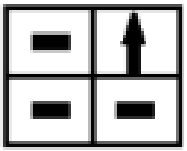
A



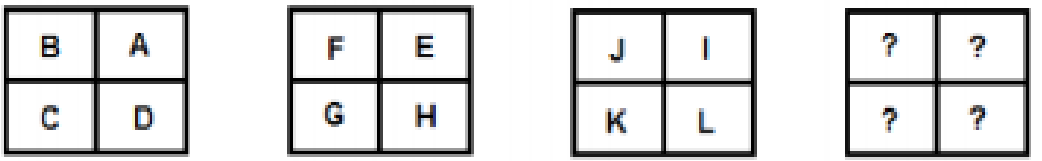
B



C



29.



A

O	M
N	Ñ

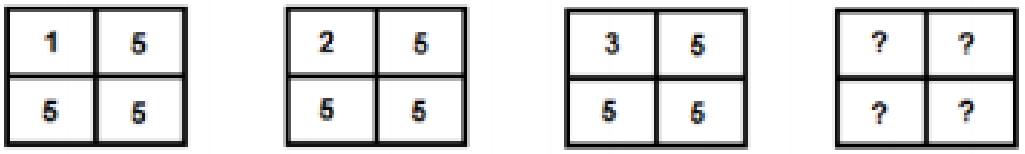
B

N	M
Ñ	O

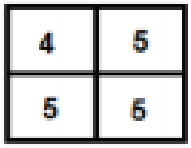
C

O	M
Ñ	N

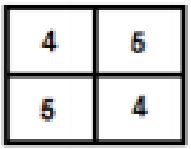
30.



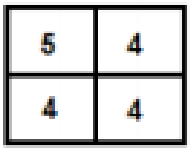
A



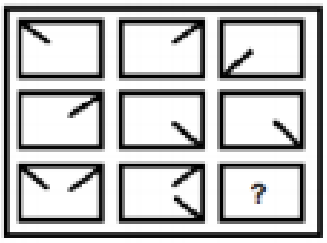
B



C



31.

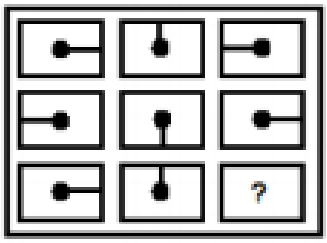


A 

B 

C 

32.

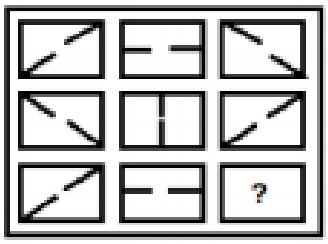


A 

B 

C 

33.

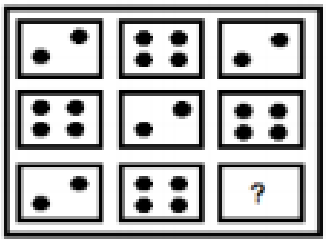


A 

B 

C 

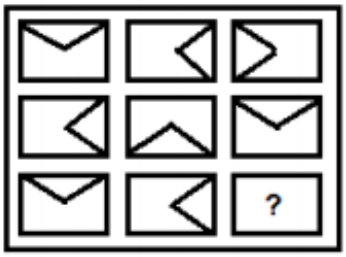
34.



A 

B 

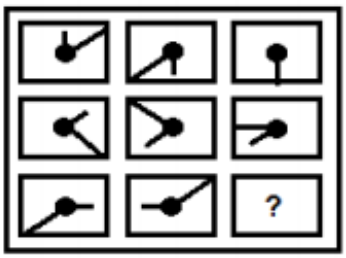
C 

35. 

A 

B 

C 

36. 

A 

B 

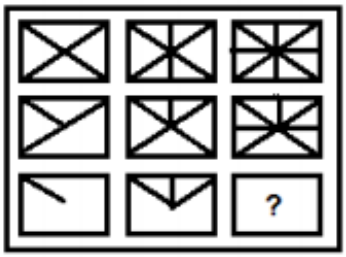
C 

37. 

A 

B 

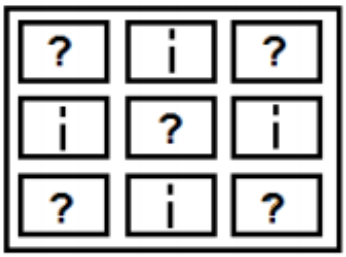
C 

38. 

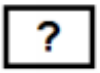
A 

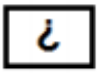
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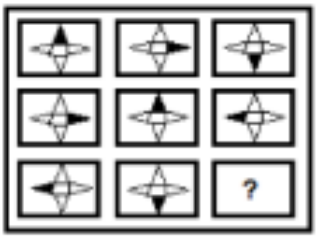
C 

39. 

A 

B 

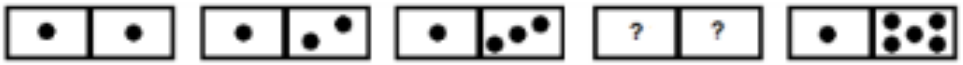
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40. 

A 

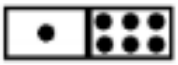
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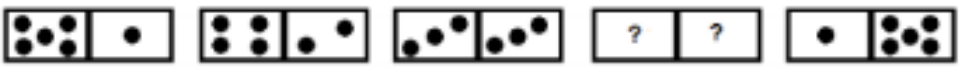
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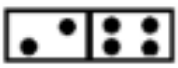
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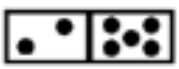
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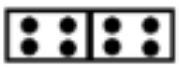
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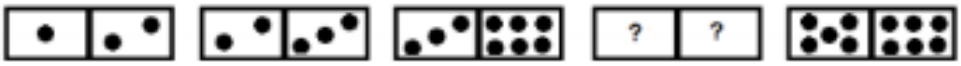
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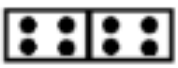
A 

B 

C 

43. 

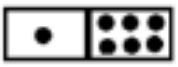
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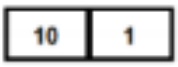
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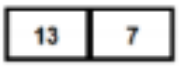
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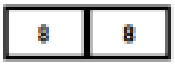
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B 

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46. 

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C 

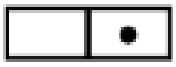
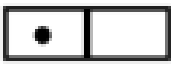
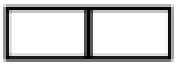
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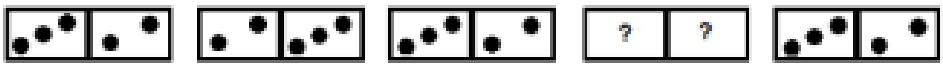
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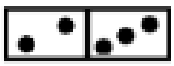
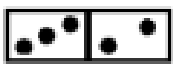
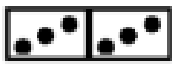
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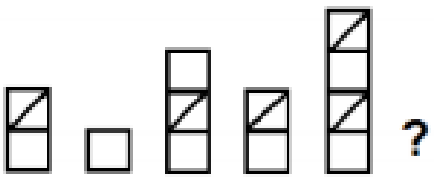
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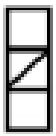
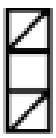
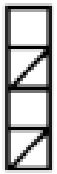
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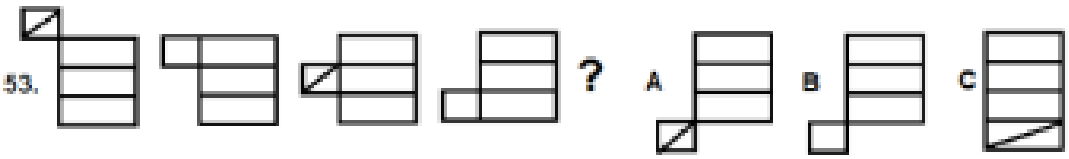
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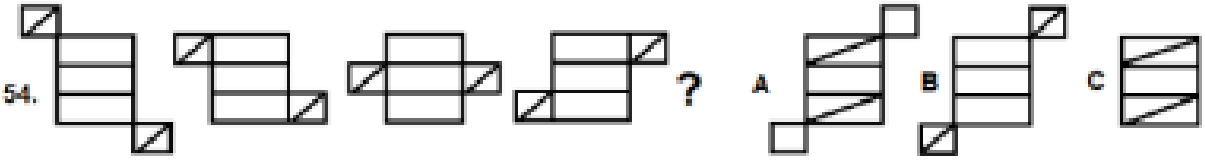
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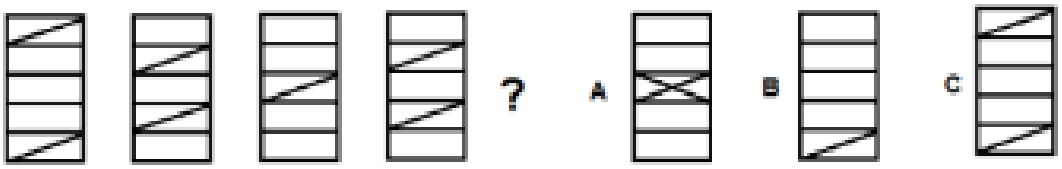
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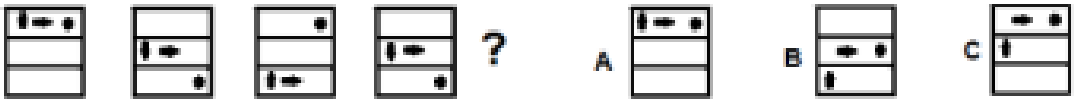
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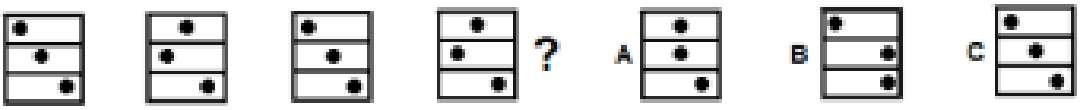
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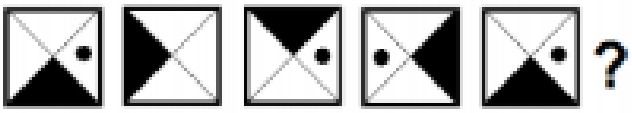
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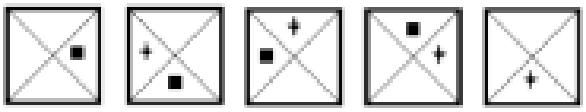
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